Stella Academy

Learn the physics and maths it takes to get to space

What we have built, where it is going, and how teachers and partners can shape it.

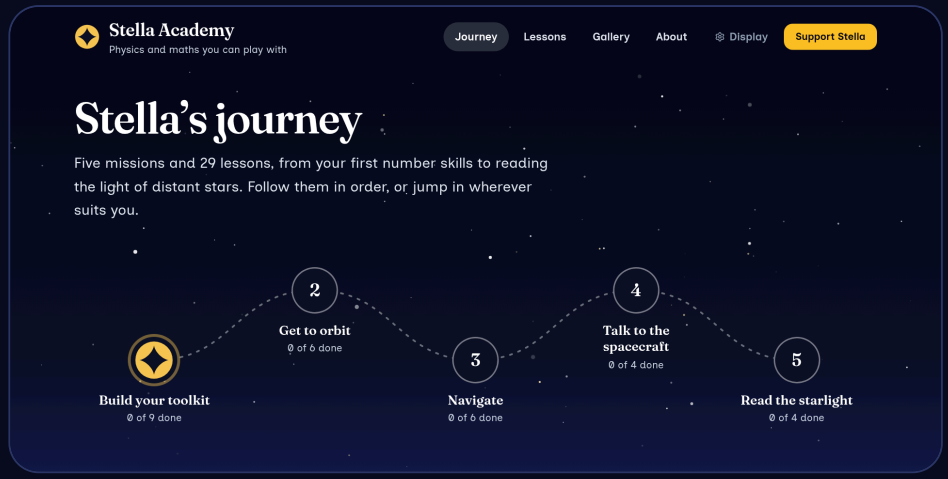
• The vision, October 2026 · www.stella-academy.org

Most children meet physics as a list of topics. We give it a destination.

Stella Academy teaches physics and maths as one journey: from first number skills to reading the light of distant stars. Every lesson answers a question a real space mission has to answer.

How fast must a rocket go to stay up? Why does a spacecraft need trigonometry to know where it is? How does a signal cross the solar system? What can a rainbow tell you about a star?

The syllabus is the same one schools already teach. What changes is the reason to care about it.



Stella's journey, as a learner sees it: five missions in order.

Five missions, 29 lessons

For ages 9 and up. Each mission is a short run of lessons in order, and each lesson takes 10 to 20 minutes.

1

Build your toolkit

The number skills every other mission relies on.

9 lessons

2

Get to orbit

What it takes to leave the ground and stay up.

6 lessons

3

Navigate

The maths of knowing where you are and where you are heading.

6 lessons

4

Talk to the spacecraft

How a spacecraft is powered, and how it sends word home.

4 lessons

5

Read the starlight

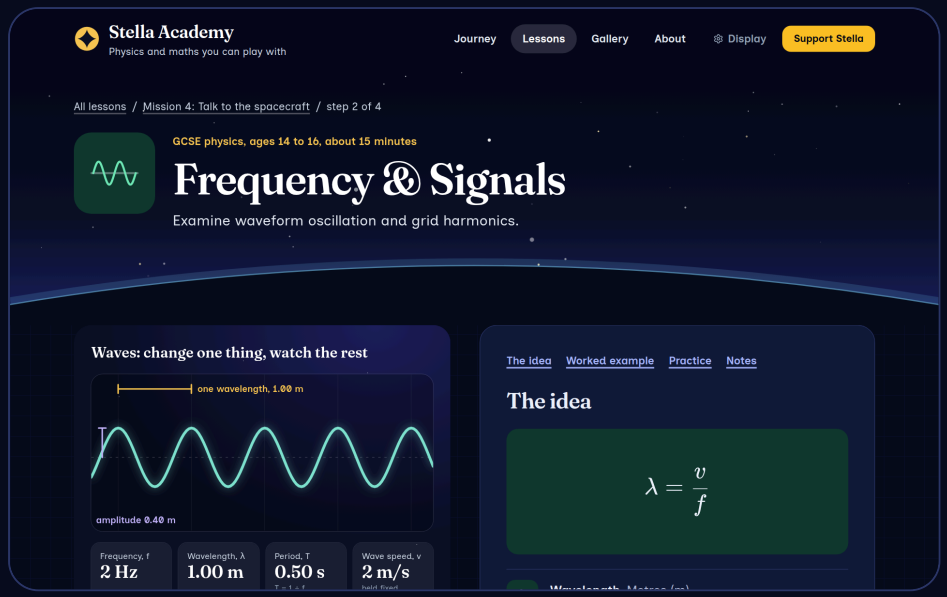
What light and data tell us about things too far away to visit.

4 lessons

Lessons come in three levels: Foundations (ages 9 to 13), GCSE (14 to 16) and A-level (16 and up). The levels are an editorial guide to difficulty. They are not yet mapped to any exam board's specification.

Learners tick lessons off as they go, and the star moves along the route. Progress is kept in their own browser; there are no accounts.

Every idea is something you can try



A GCSE lesson on waves. The model on the left responds as you change it.

A model to play with

Change one thing and watch the rest: a wave, a circuit, a prism, an orbit.

A short explanation

The idea and its formula, in plain words, with the maths set properly.

A worked example

One problem solved step by step.

Practice questions

With hints and answers to check against.

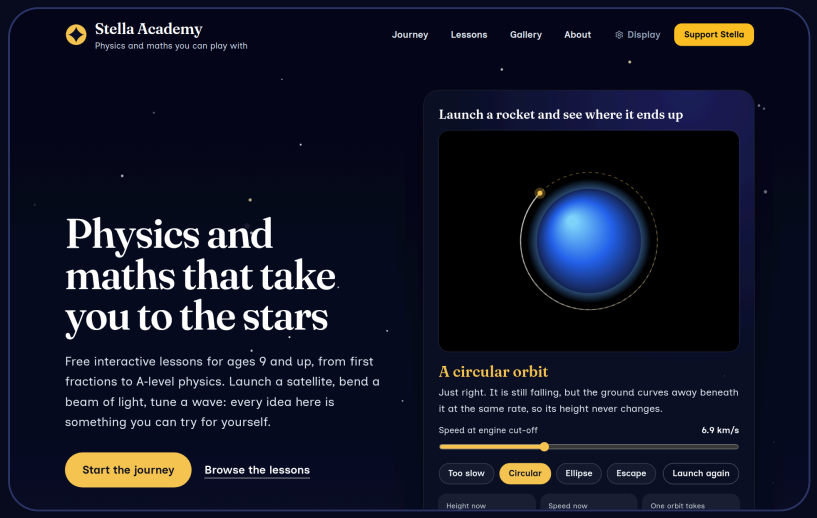
Real footage where it helps

Eight short NASA clips, each credited: a launch, a booster separating, parachutes opening, a rover driving on Mars.

Small on purpose

We are not trying to be Brilliant, Bitesize or Isaac Physics. They cover everything, with teams and budgets to match. We cover one theme and try to do it well.

That makes Stella a companion to what schools already use: the thing a teacher shows at the start of a topic, or a parent hands to a child who has asked how rockets work.



The home page: choose a speed, launch, and see whether you reach orbit.

Where things stand today

Live and free to use

29 lessons at www.stella-academy.org, open to anyone.

Drafted with AI, then checked

Every lesson has been audited for arithmetic and for errors of explanation, and corrected. Each lesson says so.

Not yet reviewed by a teacher

This is the biggest gap, and the main reason for this document.

Run by one person

Independent, with no company or charity behind it yet. Not part of NASA and not endorsed by it.

Where it goes from here

Slowly, and only as far as each step earns. Nothing is built until the step before it has shown it is wanted.

Step	What happens	We move on when
1. Written with teachers	Teachers review the lessons at their level. A reviewed lesson names its reviewer; the rest keep the AI notice.	lessons carry a teacher's name
2. Used in a classroom	One mission tried with a class or club, with a pack to go with it: lesson plan, worksheet, slides.	a teacher would use it again
3. Somewhere to keep progress	Optional accounts, created by an adult for younger children, so progress follows a learner between devices.	families ask for it
4. Paying its own way	The lessons stay free. Income comes from voluntary support, teaching packs, and later a paid tier for saved progress, marking and a tutor.	it covers its costs

We charge only for our own work. NASA's photographs and footage stay in the free lessons, credited to their source.

How you can help shape it

If you teach physics or maths

Read two or three lessons at your level and tell us what is wrong, what is pitched badly and what is missing. About 20 minutes a lesson. Your name goes on the lessons you review, if you want it there.

If you run a class or a club

Try one mission with your group and tell us what happened. “Get to orbit” is six lessons and ends with a launch simulator.

If you work in outreach

We are looking for a university or school partner to apply with for public-engagement funding, with Stella as the delivery partner.

What you get back

A named credit, a say in what is built next, and every teaching pack free. There is no budget for fees yet; review time is the first thing a grant would pay for.

Start here

www.stella-academy.org

Try the journey at /journey, then reply to whoever sent you this.

Stella Academy is an independent project. It is not part of NASA and is not endorsed by it.